

GLCM Based Feature Extraction and Medical X-RAY Image Classification using Machine Learning Techniques

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Abstract— The machine learning and artificial intelligence play a vital role to solve the challenging issues in Clinical imaging. The machine learning and artificial intelligence ease the daily life of both medical practitioner and patient's. Nowadays, the automatic system is designed with high accuracy to perceive abnormality in bone X-ray images. To achieve high accuracy system has less resource available image pre-processing tools are used to enhance the medical images quality. The image pre-processing involves the process like noise removal and contrast enhancement which provides instantaneous abnormality diagnosis system. The Gray Level Co-occurrence Matrix (GLCM) texture features are widely used in image classification problems. GLCM represents the second-order statistical information of gray levels between neighboring pixels in an image[1]. In the paper, we implemented different machine learning approaches to classify the bone X-ray images of MURA (musculoskeletal radiographs) dataset into fractures and no fracture category. The four different classifiers LBF SVM (Radial Basis Function support vector machine), linear SVM, Logistic Regression and Decision tree are used for abnormality detection. The performance evaluation of the above abnormality detection in X-ray images is performed by using five statistical parameters such as Sensitivity, Specificity, Precision, Accuracy and F1 Score, which shows significant improvement.

Keywords—Machine Learning; GLCM, LBF SVM, Linear SVM, Logistic Regression, Decision Tree, MURA, Bone Fractures

I. INTRODUCTION

Machine learning and artificial intelligence are application areas for the future of clinical imaging. Some devices such as X-rays, magnetic resonance imaging (MRI) and computed tomography (CT) help clinicians to quickly identify the various abnormalities in the clinical imaging. Some of the advanced features are used to detect that humans cannot identify without applying machine learning and artificial intelligence. The new emerging era in the form of

intelligent system support for machines provides radiologists with enormous support in clinical workflows.

A. Computer-Aided Detection (CAD) systems

The Medical image analysis is a tough task for physicians to make ample use of their skill, knowledge and imaging techniques. The automatic system for abnormality discovery in the clinical X-ray image is a challenging problem in the machine learning field. There are expansive anatomical varieties for each patient. Therefore, this is one of the most critical issues in radiographs projection, which contains superimposed structures.

The Radiologist implies their skill and practical knowledge to examination X-ray image for the fracture detection in the bone. The quality of the clinical image captured from image capture tool is of poor quality, so it becomes hard for the medical practitioner to detect the abnormality, an efficient automatic computer-aided detection system is designed to overcome the problem. Thus, the automatic abnormality detection of X-ray image helps the radiologist to diagnose several clinical issues such as arthritis, dental decay, bone cancer, osteoporosis, fracture and infection.

B. Musculoskeletal Radiographs (MURA)

A musculoskeletal radiograph (MURA) is a massive dataset of bone X-ray. The calculations are entrusted to decide that X-ray images are normal or abnormal. Musculoskeletal conditions influence more than two billion individuals around the world and most widely recognized reason for serious, long haul agony and inability with 30 million crisis division visits yearly. MURA dataset can prompt critical advancement in medicinal imaging which can analyse the dimension of specialists towards changing social insurance. MURA is one of the most significant open radiographic normal and abnormal X-ray dataset contains total no of 36770 images out of which 20828 are normal and 15942 abnormal X-ray images, i.e. 56.64% normal data and 43.36% abnormal data.

II. RELATED WORK

K. Haralick et al. [2] introduced texture features using GLCM way back in 1973. These texture features have been extensively used in image classification applications, most common in the area of biomedical. H. Chai et al. [3] proposed the methodology to evaluate the performance of the femur's long bone fracture using GLCM in the year 2011. S. Mahendran et al. [4] implemented fusion-classification technique to detect long bone fracture in tibia bones in particular (Leg Bones) in year 2011. V. Bousson et al. [5] design a clinical tool to calculate trabecular bone score and bone mineral density to determine the quality of bone through pixel gray-level variations in DXA images of the lumbar spine in the year 2011. R. Vijayakumar et al. [6] proposed a hierarchical algorithm for automatic classification of pelvic bone fracture via X-ray images in the year 2013. D. Yadav et al. [7] Implemented edge identification of bone X-ray images using Sobel, canny and log edge detection in the year 2011. M. Al-Ayyoub et al. [8] addressed the problem to determine fracture in long bone and classification of fracture type with the help of different classification algorithm in the year 2013. H. Mahmoud et al. [9] proposed an efficient system for detection of hand bone fractures by applying noise filter and features (Wavelet and Curvelet transform) extraction using labeled X-ray Images in year 2013. S. Srivastava et al. [10] introduced a system to identify breast cancer by applying quantum noise reduction, contrast enhancement (CLAHE). Total eighty-eight hybrid features are extracted and mammogram images are examined using an unsupervised genetic algorithm (GA), and forward sequential feature selection (SFS) in year 2014. M. Nagarajan et al. [11] proposed a system to evaluate fracture risk estimation using bone mineral density by computation trabecular bone characterizing feature set in the year 2015. V. Ngo et al. [12] characterized Bone Texture based on Gabor Transformation and Contourlet and compared HSCF performance against haralick, Gabor filter and SFTA Algorithm in the year 2016. M. Shirvaikar et al. [13] proposed a statistical method to enhance the prediction of bone fragility with the help of bone architecture and bone mineral density via GLCM Textural Features in the year 2016. Rajpurkar et al. [14] proposed a 169-layer descent baseline model to diagnose abnormality in Musculoskeletal Radiographs X-ray images in the year 2017. J. Ding et al. [15] implemented several machine learning-based algorithms such as SVM, K-mean clustering and deep learning approach, defined by seventeen GLCM feature which was extracted for verification and validation of Massive Scale Image Data in the year 2017. C. Xia et al. [16] implemented support vector machine-based bone tumor identification via extracting the GLCM based texture features of X-ray image with 5-fold cross-validation test in the year 2018. Z. Xing et al. [17] proposed effected salap swarm algorithm to optimized GLCM with fitness function and diagonal class entropy experimented on the natural satellite image in the year 2019.

III. PROPOSED WORK

The proposed method consists of four major components, namely:

- 1) Image Pre-processing,
- 2) Feature Extraction,
- 3) Splitting Dataset,
- 4) Classification Model.

The block layout of our proposed methodology is described below in Figure. 1.

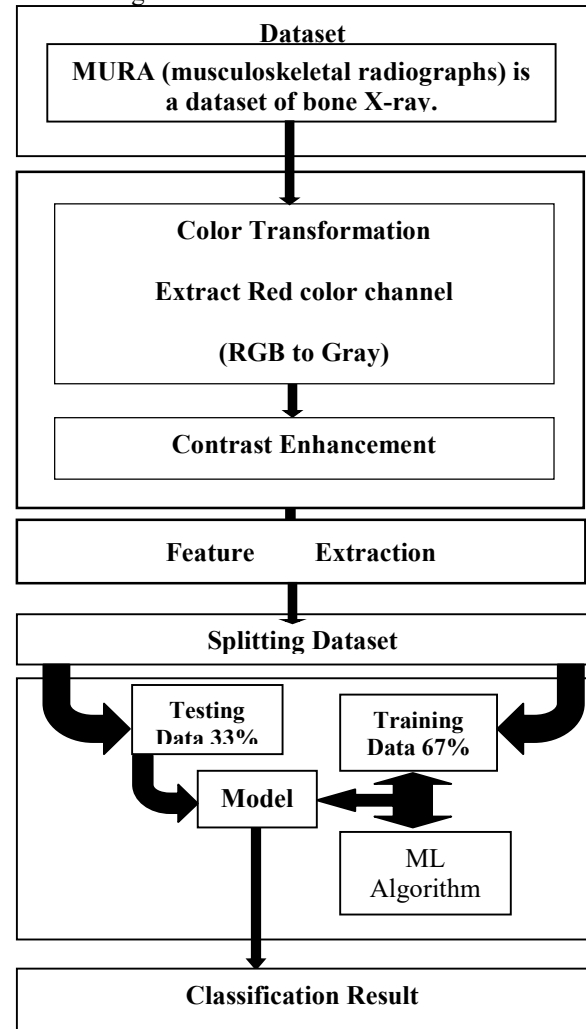


Figure 1: Block Layout of our proposed Methodology

A. Image Preprocessing

In Pre-processing phase, the clinical image is primarily processed to minimize the distortion created via noise interference and improve the vital information of the original image. The main aim is to implement color transformation technique like RGB to Gray, extract red layer information and enhanced contrast of image etc.

- *Color transformation:* In the preprocessing step, it is required to convert the RGB image into a grayscale image, which can be achieved using the average method or luminosity method. In proposed methodology

luminosity method is used in which 100% of red, 0% of green, 0% of the blue portion is used.

- **Contrast Enhancements:** The most important phase of the preprocessing is to reduce unwanted distortion signal (noise) and enhance the contrast in the image. The X-ray image captured by the imaging modalities device may be imperfect and lacking in some valuable information such as uneven staining and low contrast; thus, it is required to enhance the object in the image. The CLAHE (contrast limited adaptive histogram equalization) technique is implemented to enhance X-ray images contrast quality. Shown in Figure 2 and 3 original and improved X-ray images, respectively.



Figure: 2 Image before enhancement

B. Feature Extraction

Feature extraction of an image is an important part of our proposed model. GLCM texture feature works on the phenomena of the co-occurrence of neighbouring gray level and its counts in the image. The GLCM texture feature is calculated using a square matrix based on ROI (region of interest) dimension of the number of gray levels (N) in the X-ray images. In this paper total twelve, numbers of the features were extracted; detail of each feature is given below.



Figure: 3 Image after enhancement

Let us assume M co-occurrence matrix with N dimension and i, j are its coefficients and coordinates of the elements.

- **Autocorrelation:** The autocorrelation is the measurement of roughness and fitness present in the

image as a texture feature. This feature is associated with the dimension of the texture primal as the fitness of the texture. For normal or smooth textures in an X-ray image, the autocorrelation method shows peaks and valleys. The autocorrelation feature of an image $I(x, y)$ is defined in Equation (1) as follows:

$$P(x, y) = \frac{\sum_{u=0}^N \sum_{v=0}^N I(u, v) I(u+x, v+y)}{\sum_{u=0}^{N_g-1} \sum_{v=0}^{N_g-1} I^2(u, v)} \quad (1)$$

- **Contrast:** The contrast is a measure of the spatial frequency of image, i.e. pixel intensity and its neighbour over the image. It also determines the amount of local variations present in the image. The contrast is measured by the difference in the color and brightness of each object with other objects within the same field of view. The contrast function of an image 'Cont.' is defined in Equation (2) as follows:

$$Cont. = \sum_i \sum_j |i - j|^2 p(i, j) \quad (2)$$

- **Correlation:** The correlation feature of an image is a measurement of how a pixel is correlated with its neighbouring pixel over the entire image. Correlation value varies from -1 or 1 for negative correlation of image perfectly positively, and value for a constant image is infinity. The Correlation feature of an image 'Correl' is defined in equation (3) as follows:

$$Correl = \sum_i \sum_j \frac{(i - \mu_i)(j - \mu_j) p(i, j)}{\sigma_i \sigma_j} \quad (3)$$

- **Cluster Prominence:** The cluster prominence characterizes the propensity of clustering of the pixels in the ROI. It measures the asymmetry of the GLCM. A higher value indicates more asymmetry, whereas a lower value points out the peak of the distribution centred around the mean. The cluster prominence feature of an image 'Pro' is defined in Equation (4) as follows:

$$Pro = \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} (i + j - u_x - u_y)^4 p(i, j) \quad (4)$$

- **Cluster shade:** The cluster shade feature measures skewness of the GLCM matrix, and it is a concept of uniformity in the image. A higher value indicates asymmetry. The cluster shade feature of an image 'Sha' is defined in equation (5) as follows:

$$Sha = \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} (i + j - u_x - u_y)^3 p(i, j) \quad (5)$$

- **Dissimilarity:** The dissimilarity feature measures the distance between pairs of objects in the ROI. It measures the gray level mean difference in distribution of the image. A larger value implies a greater disparity in intensity values among neighbouring pixels. The dissimilarity feature of an image 'Dissimilarity' is defined in equation (6) as follows:

$$Dissimilarity = \sum_i \sum_j |i - j| p(i, j) \quad (6)$$

- **Energy:** The energy feature measures the degree of pixel pair repetitions. It is the measurement of disorder in texture in an image. With highly correlated pixels, the energy value is very large. The energy feature of an image 'Energy' is defined in equation (7) as follows:

$$\text{Energy} = \sum_{i,j} p(i,j)^2 \quad (7)$$

- **Entropy:** The entropy feature measures the degree of uniformity between pixel within the image and randomness is used to characterize the texture of the image. Entropy is strongly but inversely correlated to energy as images with a larger gray levels has high entropy. The entropy function of an image 'Ent' is defined in equation (8) as follows:

$$\text{Ent} = \sum_{i=0}^{N_g-1} \sum_{j=0}^{N_g-1} p(i,j) \log(p(i,j)) \quad (8)$$

- **Homogeneity:** The homogeneity is commonly referred to as inverse different movement. It measures image homogeneity with larger values for smaller gray tone difference in pair object. Homogeneity decreases as contrast increases with constant energy. The homogeneity feature of an image 'Homog' is defined in equation (9) as follows:

$$\text{Homog} = \sum_i \sum_j \frac{1}{1+|i-j|^2} P(i,j) \quad (9)$$

- **Maximum probability:** The maximum probability measures the maximum likelihood of producing the pixels of interest. The maximum probability feature of an image 'Max. Probability' is defined in equation (10) as follows:

$$\text{Max. Probability} = \max. p(i,j) \text{ for all } (i,j) \quad (10)$$

- **Sum of squares variance:** The sum of squares variance measures the dispersion of the gray level sum distribution of the image. This heterogeneity is strongly correlated to a first-order statistical variable, i.e. standard deviation. As the gray level value differs from its mean value, there is an increase in the variance of image. The sum of squares variance feature of an image 'Variance' is defined in equation (11) as follows:

$$\text{Variance} = \sum_{i=1}^{N_g} (i - \mu)^2 p(i,j) \quad (11)$$

- **Sum average:** The sum average feature measures the mean of the gray level sum distribution of an image. The sum average feature of an image 'Sum average' is defined in equation (12) as follows:

$$\text{sum average} = \sum_{i=2}^{2N_g} ip_{x+y(i)} \quad (12)$$

IV. SIMULATION AND RESULTS

The simulation of the proposed methodology is categorized into three major parts: image preprocessing, feature extraction and classification. First two parts image preprocessing, and feature extraction has been performed using MATLAB R2017a environment, and last part classification has been performed on python 3.0.

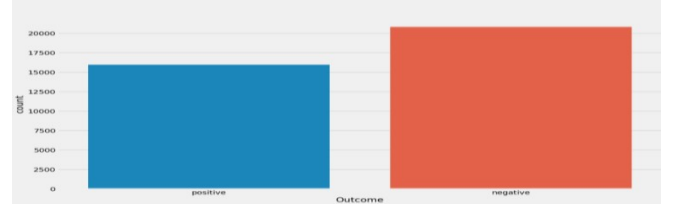


Figure 4: Total number of count positive and negative records

The above bar graph (Figure 4) indicates the total number of positive and negative records of the MURA database, which contains total no of 36770 X-ray images out of which 15942 positive images and 20828 negative images. The proposed methodology is implemented using it.

In the proposed method total, twelve number of features were extracted from an X-ray image of MURA dataset using a GLCM, above histograms (Figure 5) displays the most repeated value of each feature extracted. Features like autocorrelation, contrast, cluster prominence, cluster shade, sum of squares variance and sum average values are very less scattered, some other features like correlation, dissimilarity, entropy, energy, maximum probability and homogeneity values are highly scattered. Heat map of twelve GLCM texture features displayed in (Figure 6) is the graphical representation of co-occurrence matrix using colour code, dark green color represent highly co-related feature and red color represent features with less co-relation.

The performance analyses of different classification algorithm are represented in the bar graph (Figure 7). Pre-processing is worked for the removal of noise and improving the quality of the X-ray image.

Through GLCM, following textural features are extracted autocorrelation, contrast, correlation, cluster prominence, cluster shade, dissimilarity, energy, entropy, homogeneity, maximum probability, sum of squares variance, and sum average is used for the classification of the X-ray images.

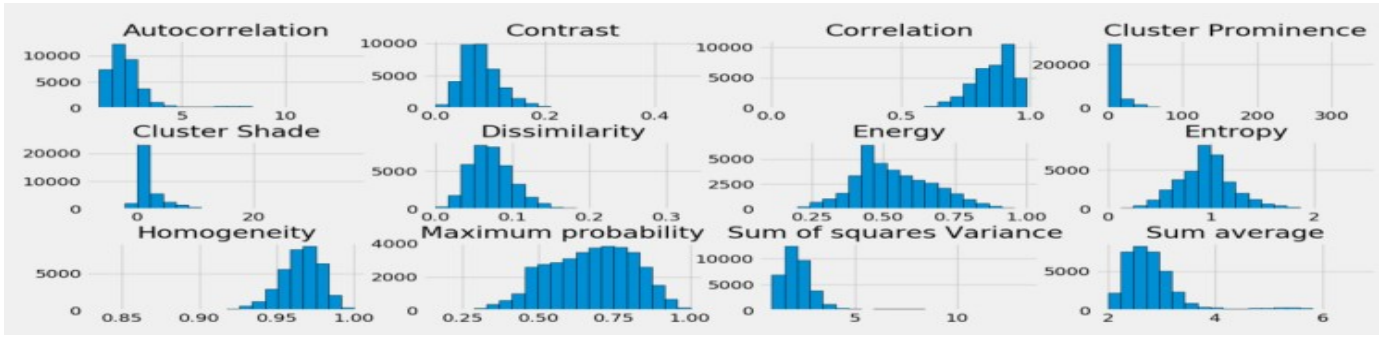


Figure 5: Extracted feature value count

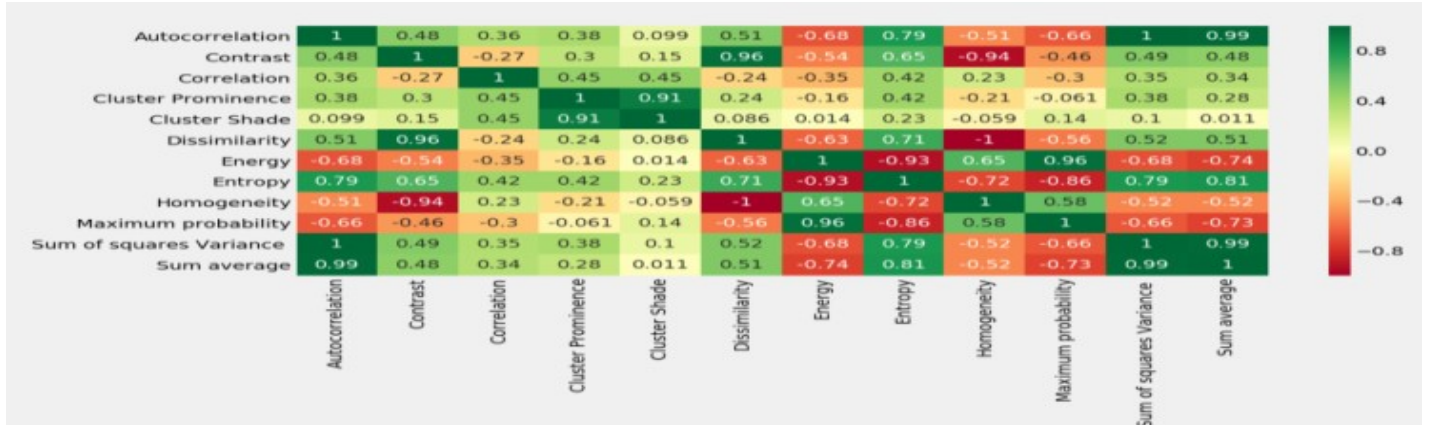


Figure 6: Heat map of twelve extracted feature

As the number of textural features increases, the performance of classifier becomes better, but it increases the computational complexity. By analysing the statistical parameters, Sensitivity, Specificity, Precision, Accuracy, and F1 Score, the classifier's performance can be evaluated. Through the proposed work, a radiologist could better identify the fractures of bone X-ray. The main objective of our proposed work is performance evaluation through different classifier algorithms with the outcomes of different textural features obtained from GLCM. One of the important steps is training the classification algorithm for which 67 % of MURA dataset and the remaining 33% dataset are used for testing purpose.

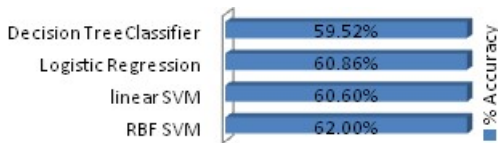


Figure 7: Analysis of different classification algorithms accuracy using texture features GLCM

Table 1: Confusion matrix of normal and fractures bone X-ray images classification (i) LBF SVM (ii) linear SVM (iii) Logistic regression (iv) Decision tree

Actual	Yes	No
Actual No	1215	5659

(i)

Actual	Yes	No
Actual No	865	6009

(ii)

n=12135	Predicted Yes	Predicted No
Actual Yes	1926	3335
Actual No	1415	5459

(iii)

n=12135	Predicted Yes	Predicted No
Actual Yes	2866	2395
Actual No	2866	4357

(iv)

The performance evaluation of this abnormality detection in X-ray images is done using five statistical parameters such as sensitivity, specificity, precision, accuracy, and F1 score.

$$\text{Sensitivity} \quad \text{TPR} = \text{TP} / (\text{TP} + \text{FN}) \quad (13)$$

$$\text{Specificity} \quad \text{SPC} = \text{TN} / (\text{FP} + \text{TN}) \quad (14)$$

$$\text{Precision} \quad \text{PPV} = \text{TP} / (\text{TP} + \text{FP}) \quad (15)$$

$$\text{Accuracy} \quad \text{ACC} = (\text{TP} + \text{TN}) / \text{Total Number} \quad (16)$$

$$\text{F1 Score} \quad \text{F1} = 2\text{TP} / (2\text{TP} + \text{FP} + \text{FN}) \quad (17)$$

Where true positive (TP) is referred to those X-ray images that are positively labelled as fractured and classified as same, true negative (TN) is referred to those X-ray images that are positively labelled as non-fractured and classified as same. False positive (FP) is referred to the X-ray images that are

n=12135	Predicted Yes	Predicted No
Actual Yes	1865	3396

n=12135	Predicted Yes	Predicted No
Actual Yes	1345	3916

labelled as non-fractured but classified as fracture image. False-negative (FN) is referred to those X-ray images that are labelled as fractured but classified as non-fracture image. For the classification of abnormality of the fracture X-ray image, several machine learning algorithm were implemented such as LBF SVM, linear SVM, logistic regression and decision tree. Above statistical performance, the measure is considered to evaluate machine learning algorithm. Sensitivity commonly known as true positive rate, specificity is the measurement of false positive rate, precision is the prediction of positive observation, accuracy correct prediction observation to the total observations, F1 Score is the measurement of a weighted average of precision and sensitivity.[18] These five statistical performance values were evaluated via confusion matrix, shown in Table 1 using different machine learning technique and performance result is shown in Table 2.

Table 2: Performance evaluation of different machine learning technique.

Measure	LBF SVM	linear SVM	Logistic Regression	Decision tree
Sensitivity	0.3545	0.2557	0.3661	0.5448
Specificity	0.8232	0.8742	0.7942	0.6338
Precision	0.6055	0.6086	0.5765	0.5324
Accuracy	0.62	0.606	0.6086	0.5952
F1 Score	0.4472	0.3601	0.4478	0.5385

Based on the confusion matrix of various classification algorithms, we have calculated the performance measures, shown in Table 2; the LBF SVM classifier achieved a classification accuracy of 62.00% with sensitivity of 35.45%, specificity of 82.32%, precision of 60.55% and F1 score of 44.72, linear SVM classifier achieved a classification accuracy of 60.60% with sensitivity of 25.57%, specificity of 87.42%, precision of 60.86% and F1 score of 36.01; Logistic Regression classifier achieved a classification accuracy of 60.86% with sensitivity of 36.61%, specificity of 79.42%, precision of 60.86% and F1 score of 44.78; Decision tree classifier achieved a classification accuracy of 59.52% with sensitivity of 54.48%, specificity of 63.38%, precision of 53.24% and F1 score of 53.85%. However, among the four models assessed; so, due to the visual challenges of MURA database fracture detection LBF SVM shows significant improvement in terms of performance is encouraging.

V. CONCLUSION AND FUTURE SCOPE

The problem undertaken in this paper is abnormality detection of the Bone fracture X-ray image. We have used the dataset from MURA of musculoskeletal radiographs provided by <http://stanfordmlgroup.github.io/competitions/mura>. From medical practitioners and patient point of view, the fully automatic detection system is used for the classification of fractures. MURA is a large dataset of X-ray images with poor quality and less clarity which makes classification a difficult problem. The approach implemented to solve this issue, and the problem is subdivided into three parts pre-processing feature extraction and classification. Image pre-processing phase is categorized as RGB to gray level conversion, and

CLAHE is used for contrast enhancement. 12 GLCM texture features are extracted and further used for image abnormality detection in the X-ray image. Several machine learning algorithm considered for the evaluation of automatic detection system is as follow LBF SVM, Linear SVM, logistic regression and decision tree. Above performance, evaluation is considered to evaluate machine learning algorithm. LBF SVM with an accuracy of 62.00%, gives the best result among the other machine learning algorithm. However, there is still the chance of improvement in performance for experiment with different feature sets.

ACKNOWLEDGMENT

In this paper, we have used a dataset from MURA, a large dataset of musculoskeletal radiographs: <http://stanfordmlgroup.github.io/competitions/mura> for the research and non-commercial purposes, we hope that they can share and give us the rights to use it.

REFERENCES

- [1] A. Seal, D. Bhattacharjee, and M. Nasipuri, "Predictive and probabilistic model for cancer detection using computer tomography images," *Multimed. Tools Appl.*, vol. 77, no. 3, pp. 3991–4010, 2018.
- [2] K. Shanmuga. Haralick, Robert M., "HaralickShanmugamDinstein1973." 1973.
- [3] H. Y. Chai, L. K. Wee, T. T. Swee, S. Salleh, and A. K. Ariff, "Gray-Level Co-occurrence Matrix Bone Fracture Detection Center for Biomedical Engineering Biomedical Engineering Group," vol. 8, no. 1, pp. 26–32, 2011.
- [4] S. K. Mahendran and S. S. Baboo, "An Enhanced Tibia Fracture Detection Tool Using Image Processing and Classification Fusion Techniques in X-Ray Images," *Glob. J. Comput. Sci. Technol.*, vol. 11, no. 14, pp. 22–28, 2011.
- [5] V. Bousson, C. Bergot, B. Sutter, P. Levitz, and B. Cortet, "Trabecular bone score (TBS): Available knowledge, clinical relevance, and future prospects," *Osteoporos. Int.*, vol. 23, no. 5, pp. 1489–1501, 2012.
- [6] R. Vijayakumar and G. Gireesh, "Quantitative analysis and fracture detection of pelvic bone X-ray images," *2013 4th Int. Conf. Comput. Commun. Netw. Technol. ICCCNT, 2013*, 2013.
- [7] Madhulika et al., "Implementing edge detection for medical diagnosis of a bone in Matlab," *Proc. - 5th Int. Conf. Comput. Intell. Commun. Networks, CICN, 2013*, pp. 270–274, 2013.
- [8] M. Al-Ayyoub and D. Al-Zghool, "Determining the type of long bone fractures in x-Ray images," *WSEAS Trans. Inf. Sci. Appl.*, vol. 10, no. 8, pp. 261–270, 2013.
- [9] H. R. Mahmoud Al-Ayyoub, Ismail Hmeidi, "Detecting Hand Bone Fractures in X-Ray Images," *Jmpt*, vol. 4, no. 3, pp. 155–168, 2013.
- [10] S. Srivastava, N. Sharma, S. K. Singh, and R. Srivastava, "Quantitative Analysis of a General Framework of a CAD Tool for Breast Cancer Detection from Mammograms," *J. Med. Imaging Heal. Informatics*, vol. 4, no. 5, pp. 654–674, 2014.
- [11] M. B. Nagarajan et al., "Characterizing trabecular bone structure for assessing vertebral fracture risk on volumetric quantitative computed tomography," *Med. Imaging, 2015, Biomed. Appl. Mol. Struct. Funct. Imaging*, vol. 9417, p. 94171E, 2015.
- [12] V. Q. Ngo and T. N. Dinh, "Bone Texture Characterization Based on Contourlet and Gabor Transforms," *Int. J. Comput. Theory Eng.*, vol. 8, no. 2, pp. 177–181, 2016.
- [13] M. Shirvaikar, N. Huang, and X. N. Dong, "The Measurement of Bone Quality Using Gray Level Co-Occurrence Matrix Textural Features," *J. Med. Imaging Heal. Informatics*, vol. 6, no. 6, pp. 1357–1362, 2016.
- [14] P. Rajpurkar et al., "MURA: Large Dataset for Abnormality Detection in Musculoskeletal Radiographs," Dec. 2017.
- [15] J. Ding, X.-H. Hu, and V. Gudivada, "A Machine Learning Based Framework for Verification and Validation of Massive Scale Image Data," *IEEE Trans. Big Data*, vol. 7790, no. c, pp. 1–1, 2017.
- [16] C. Xia et al., "SVM-Based Bone Tumor Detection by Using the Texture Features of X-Ray Image," *Proc. 2018 6th IEEE Int. Conf. Netw. Infrastruct. Digit. Content, IC-NIDC, 2018*, vol. 6, no. 6, pp. 130–134, 2018.
- [17] Z. Xing and H. Jia, "Multilevel Color Image Segmentation Based on GLCM and Improved Salp Swarm Algorithm," *IEEE Access*, vol. 7, pp. 37672–37690, 2019.
- [18] K. K. Sharma and A. Seal, "Modeling uncertain data using Monte Carlo integration method for clustering," *Expert Syst. Appl.*, vol. 137, pp. 100–116, 2019.